NORTHBRIDGE TECHNICAL COLLEGE

CMP 110 Programming Fundamentals

Final examination - Academic year 2020, semester 1

Time allowed: 3 hours Total marks: 100 Lecturer: Mr T. Hamoonga

Instructions: answer ALL questions. Write clearly. Show all working.

Question 1: Data structures intro

(a) Which structure stores key-value pairs?

(b) A string is essentially a sequence of:

(c) Which list operation adds an item to the end?

(d) What is the term for the number of characters in a string?

Question 2: Iteration

(a) Which loop is best when the number of iterations is known in advance?

(b) What does this code print? i = 1; while (i < 4) { print(i); i++; }

(c) A counter variable in a loop is used to:

(d) What is a loop called that never terminates?

Question 3: Programming basics

(a) Which symbol usually ends a statement in C, Java and Dart?

(b) Comments in code are:

(c) An IDE is:

(d) The entry point of a C or Java program is the function named: